

Physics Unit 7, Part 7A

Density, $\rho = m/V$, liquids aur regular solids ki density · kitab p168 se p172

Seemab, ye woh hissa hai jo tumhare scan mein chhoot gaya tha. Kitab ke chhe numericals mein se **chaar** yahin se bantay hain, to ise itminan se parhna.

1. Density (7.1, kitab p168 aur p169)

Kitab ki definition: exam mein bilkul yehi likhna

"Mass per unit volume of the substance is called its density."

Density = Mass / Volume, symbol: $\rho = m / V$ (Equation 7.1)

SI unit: **kilogram per cubic metre (kg/m^3)**. Smaller unit: **gram per cubic centimetre (g/cm^3)**.

Density of liquids is usually measured in **gram per millilitre (g/mL)**. It is a **scalar quantity**.

Matlab: ek fix volume mein kitna matter bhara hua hai. Jis cheez ke atoms jitne **closely packed** honge utni woh **denser** hogi, isi liye metals aur rocks dense hain. Kitab (Figure 7.1): "Density of solids is greater than liquids and density of liquids is greater than gases."

Do book lines: "Density of small amount of a substance is the same as the density of its bulk" (choti si cheeni aur पूरी बori, density wahi), aur "density of a material changes with the temperature" (garam karne par volume phailta hai, mass wahi, to density **kam**). **Can you tell:** liquids gases se denser hain kyunke un ke particles paas paas hain; increasing order = hydrogen, milk, mercury, gold.

Unit conversion, ye hamesha kaam aayega

$1 \text{ g/cm}^3 = 1000 \text{ kg/m}^3$. g/cm^3 ko kg/m^3 banana ho to **1000 se multiply** karo ($0.9 \times 1000 = 900$), ulta karna ho to 1000 se divide. **$1 \text{ mL} = 1 \text{ cm}^3$** , is liye 1 g/mL aur 1 g/cm^3 ek hi cheez hain.

Material	Density (kg/m^3)	Material	Density (kg/m^3)
Iron	7900	Pure water	1000
Gold	19300	Mercury	13600
Ice	920	Air	1.3
Polythene	900	Carbon dioxide	2.0
Petrol	800	Table 7.1, kitab p169	

Example 7.1: Density of regular shape (kitab p169)

Given: Mass of the material = 500 g; side length of the cube = 5 cm

Required: Density of the material $\rho = ?$

Solution: Volume of a cubic shape object = (side length)³

Volume = $(5 \text{ cm})^3 = 125 \text{ cm}^3$

$\rho = \text{Mass of cube} / \text{Volume of cube} = 500 \text{ g} / 125 \text{ cm}^3 = 4 \text{ g/cm}^3$

Kitab ki teen galtiyan: inhen yaad rakho

1. Petrol. Table 7.1 (p169) petrol **800 kg/m^3** deti hai, magar FOR YOUR INFORMATION box (p168) wahi petrol **$0.9 \text{ g/cm}^3 = 900 \text{ kg/m}^3$** bana deta hai. Dono sahi nahin ho sakte. Poocha jaye to **Table 7.1 wala 800** likhna; 0.9 se 900 wali baat sirf conversion ka method yaad rakhne ke liye hai.

2. Osmium. Do You Know box (p168) osmium ko sab se dense material kehta hai (22.59 g/cm^3), phir likh deta hai "It is harder than diamond". Ye **ghalat** hai: osmium sab se **dense** hai, hard sab se **diamond**. Dense aur hard alag cheezein hain.

3. Numerical Q5 (p188), iron cube ki cavity. Chhapa jawab **935.9 cm^3** tab hi aata hai jab iron **7.8 g/cm^3** lo, jabke Table 7.1 **7.9** deti hai, jis se **936.7 cm^3** banta hai. Dono jawab theek mane jayenge.

2. Density of liquids (7.2, kitab p170 aur p171)

7.2 sab methods ka aghaz hai: liquids, regular solids aur irregular solids. Liquid ke liye do cheezein chahiyein, **graduated measuring cylinder** (volume) aur **balance** (mass).

1. Place the empty measuring cylinder on balance and measure its mass.

2. Add liquid in the measuring cylinder and measure its volume.
3. Subtract mass of empty cylinder from the mass of cylinder and liquid:
 $\text{Mass of liquid} = \text{Mass of measuring cylinder and liquid} - \text{Mass of empty cylinder}$
4. Measure volume of liquid from the measuring cylinder.
5. To calculate the density of liquid, divide mass of liquid by its volume.

Density of liquid = Mass of liquid / Volume of liquid (Equation 7.2)

Pooru khel step 3 ka hai: balance khali cylinder ka mass bhi gin leta hai, is liye **ghatana zaroori** hai. **Figure 7.2** ke do balance: khali cylinder (145), cylinder plus liquid (205); kitab ne dono ko galti se "Step 1" likha hai, doosra **Step 2** hai.

Example 7.2: Density of liquid (kitab p171)

Given: Mass of empty cylinder = 145 g; mass of cylinder and added liquid = 205 g; volume of liquid = 60 mL

Required: Density of the liquid $\rho = ?$

Solution: Mass of liquid = 205 g – 145 g = **60 g**

$\rho = \text{Mass of liquid} / \text{Volume of liquid} = 60 \text{ g} / 60 \text{ mL} = \mathbf{1 \text{ g/mL}}$

Kitab ka natija: 1 g/mL paani (at 4 °C) ki density hai, to cylinder mein jo liquid tha woh **water** hai.

3. Density of regularly shaped solids (7.2.1, kitab p171 aur p172)

1. Find mass of the solid regular shaped object using balance.
2. Calculate the volume of object using formula according to shape of the object.
3. Finally calculate its density by using the formula.

Volume formulas: yaad karna lazmi hai

Volume of cuboid = **Length × Width × Height** · Volume of cube = **(side length)³**

Volume of sphere = **$(4/3) \pi r^3$** (r = radius, yaani diameter ka aadha) · Density of Solid = Mass of Solid / Volume of Solid (Equation 7.3)

Example 7.3: Density of brick (kitab p172, cm ko metre banane wali misaal)

Given: Mass = 3.30 kg; length 21.6 cm = 0.216 m, width 10.2 cm = 0.102 m, height 6.35 cm = 0.0635 m

Required: Density of the solid object $\rho = ?$

Solution: Volume = Length × Width × Height = 0.216 × 0.102 × 0.0635 = **0.00140 m³**

$\rho = 3.3 \text{ kg} / 0.00140 \text{ m}^3 = 2358.773 = \mathbf{2360 \text{ kg/m}^3}$

Pakki trick: units pehle milao

Jawab **kg/m³** chahiye to mass **kg** aur volume **m³** (cm ko 100 se divide kar ke metre banao, 21.6 cm = 0.216 m); jawab **g/cm³** chahiye to mass **g** aur volume **cm³**. kg ko cm³ se divide karna sab se aam galti hai.

Irregular cheez (pathar) ka volume **displacement method** se nikalta hai: woh 7.2.2 hai, Part 7B ke notes mein aa chuka hai.

Quiz

Ye teeno sawal tumhare Physics ke paper mein bhi aayenge.

1. State the book's definition of density, write Equation 7.1 with its SI unit, and then write the volume formulas the book gives for a cuboid, a cube and a sphere.
2. An empty measuring cylinder has a mass of 120 g. When 50 mL of a liquid is poured into it, the total mass becomes 160 g. Find the density of the liquid in g/mL, then convert your answer into kg/m³.
3. A small cube and a large cube are cut from the same block of iron. Why is the density of both cubes exactly the same, even though their masses and volumes are different?